NexaTel Customer Churn Analytics Project

DATA ANALYTICS PROJECT REPORT

Internship & Submission Details

Intern Name	Sugumar S	**Domain**	Data Analyst Intern
Company Name	Internmo	**Intern ID**	Inter779
Batch	August – November 2026	**Status**	Remote
Email	sugumar338@gmail.com		

Project Summary

This project analyses NexaTel subscription, voice, SMS, data-usage, customer-support and store datasets to understand observable patterns associated with subscriber status and customer experience. The work includes data profiling and quality checks, customer-level preparation, exploratory analysis, churn and usage metrics, support-ticket analysis, SQL analysis queries, and practical retention recommendations.

17.14% Observed status-based churn	**31,490** Support tickets analysed	**57.55%** Tickets marked resolved
16.04 hrs Average resolution time	**70.22%** First-contact resolution*	**6 datasets** Source files reviewed

*Among resolved tickets. The 17.14% figure is a descriptive latest-status rate among customers with known status, not a predictive or cohort-period churn rate.

Report Sections

- **1. Executive Summary** – key findings and project outcomes
- **2. Data Inventory and Preparation** – source files, data preparation and churn definition
- **3. Customer Churn and Usage Analysis** – subscription status and monthly usage trends
- **4. Customer Support Analysis** – ticket volume, resolution KPIs and churn association
- **5. Data Quality, Limitations and Interpretation** – validation issues and analytical boundaries
- **6. Recommendations and Next Steps** – retention and operational improvement actions
- **7. Suggested Internship Work Description** – concise summary of project work

Analytical tools and methods: Python, Pandas, NumPy, Excel, SQL query development, exploratory data analysis, KPI analysis and dashboard-ready data preparation.

• Customer churn baseline: 17.14% among customers with a known latest status.

• Support workload: 31,490 tickets from 15,206 customers; 57.55% are marked Resolved.

• Resolution performance: resolved tickets average 16.04 hours; 70.22% of resolved tickets are marked first-contact resolution.

• Usage movement: mean data usage falls from 42.40 GB in January to 38.42 GB in March; the median falls from 19.33 GB to 19.03 GB.

• Ticket-count association: observed churn rises from 15.6% among customers with one ticket to 29.4% among customers with 6–9 tickets; this is an association, not causation.

2. Data inventory and preparation

File	Rows	Fields	Coverage / role
subscriptions.csv	20,523	8	Plan, start/end dates, billing cycle, charge and status
usage_voice.csv	57,000	6	19,000 customers × 3 months
usage_sms.csv	57,000	5	19,000 customers × 3 months
usage_data.csv	57,000	5	19,000 customers × 3 months
support_tickets.csv	31,490	11	Historical support interactions, 2014–2026
stores.csv	2,180	7	Store master data; 68 cities, 29 states, 4 regions

Preparation method

Dates in the supplied files were interpreted as DD-MM-YYYY; usage_month was interpreted as MM-YYYY. Subscription records were sorted by customer_id, start_date and subscription_id, and the latest record per customer was selected. The three usage tables were aggregated to customer-level means/medians. Support tickets were aggregated to customer-level counts and outcomes. The stores table was profiled but not joined to customers because no customer-to-store key was supplied.

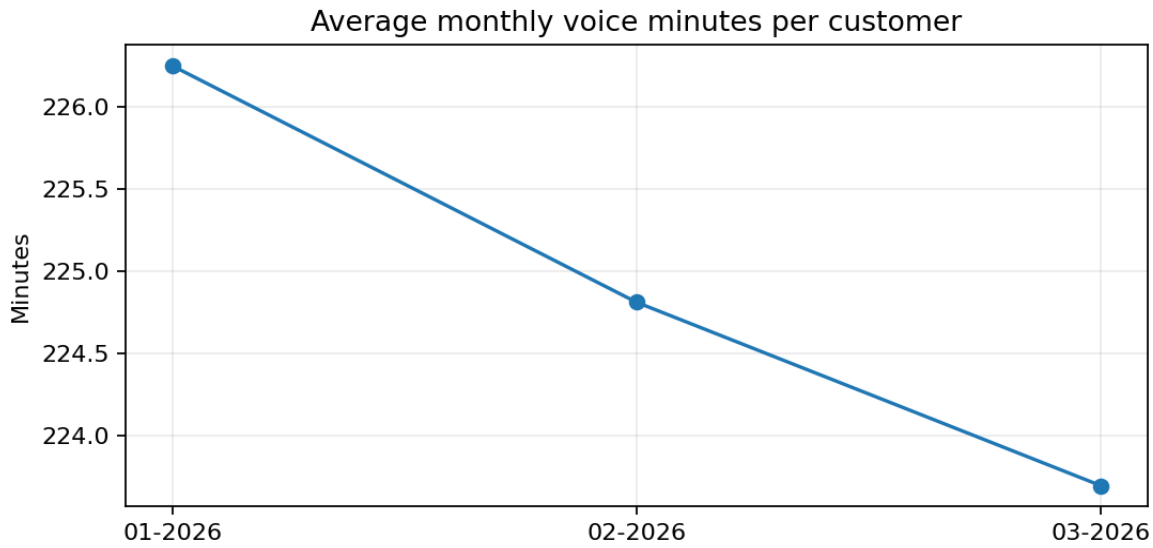
Customer churn calculation

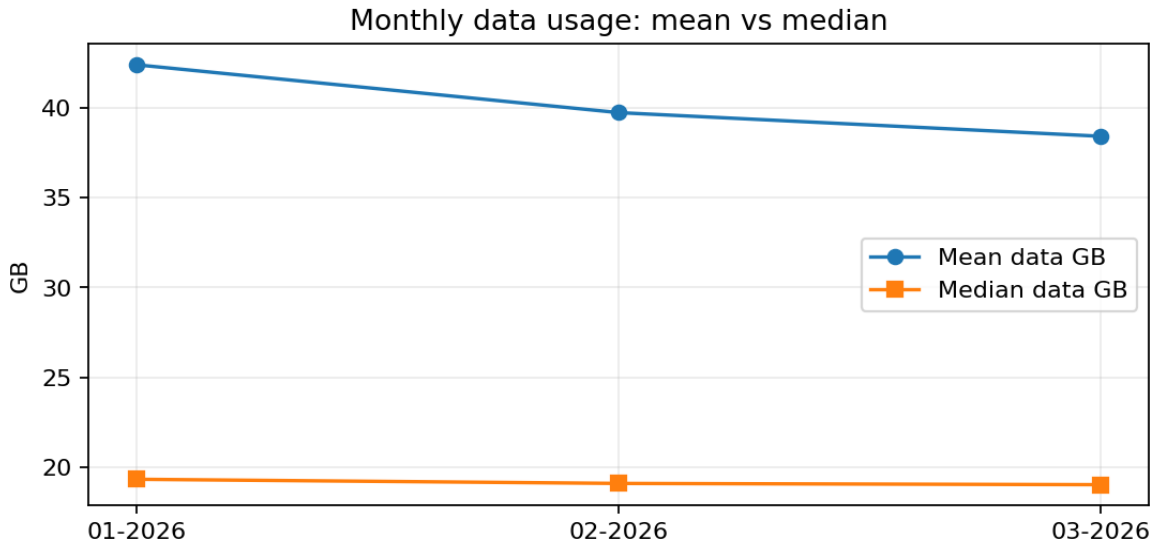
Churn flag = 1 when the latest subscription_status is Terminated; 0 when Active. UNKNOWN_STATE is excluded from the churn-rate denominator. This produces a descriptive status-based churn rate, not a cohort retention or period-specific churn rate.

3. Customer churn and usage analysis

Of 19,000 customers, 3,240 have a latest Terminated status and 15,668 have a latest Active status. 92 customers are marked UNKNOWN_STATE and should be resolved before operational use.

Usage trends





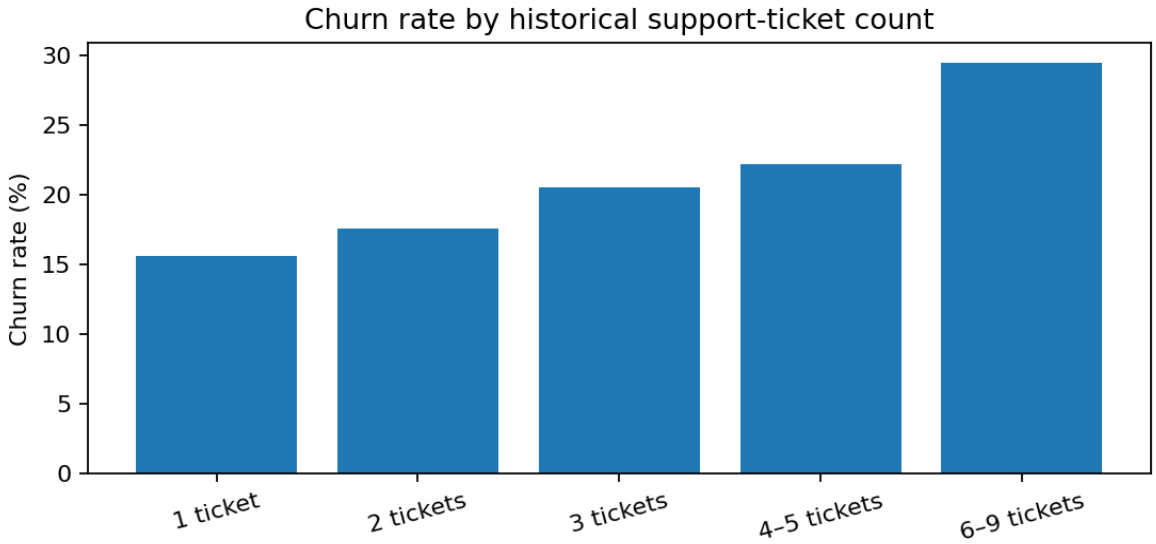
Average voice minutes decrease modestly from 226.25 in January to 223.69 in March. Mean data usage falls from 42.40 GB to 38.42 GB; the median changes from 19.33 GB to 19.03 GB. The large gap between mean and median indicates that extreme high values materially inflate average data usage.

Data-quality flag: 4,480 usage rows exceed the 1.5×IQR upper fence of 64.49 GB, including a maximum of 22,172.70 GB. These values should be validated against source units and usage logic before drawing conclusions from mean usage.

4. Customer support analysis

Support KPI	Observed result
Total tickets	31,490
Customers with at least one ticket	15,206
Resolved	18,124 (57.55%)
Pending	8,450 (26.83%)
Escalated	2,526 (8.02%)
Reopened	2,390 (7.59%)
Average resolution time (resolved)	16.04 hours
First-contact resolution (resolved)	70.22%

Ticket volume and churn association



Customers with more historical tickets show higher observed churn in this dataset. For example, churn is approximately 15.6% for customers with one ticket and 29.4% for customers with 6–9 tickets. Ticket dates span 2014–2026, while usage covers only Q1 2026; the analysis does not establish that support issues preceded or caused termination. The relationship should be validated using time-aligned ticket and subscription histories.

5. Data quality, limitations and interpretation

Items requiring validation

• Missing subscription plan: 41 of 20,523 subscription rows have a blank plan_id.

• Unresolved status coding: 102 subscription rows have UNKNOWN_STATE; at customer level this affects 92 latest records.

• Multiple subscriptions: 1523 customers have more than one subscription record. Latest-record selection was used to avoid counting customers multiple times.

• Future-dated starts: 52 subscription rows start after the usage period ends (31 March 2026), including 11 rows dated after 30 October 2026.

• Usage after termination: 2808 latest-terminated customers have an end date before January 2026, yet the usage files still contain Q1 2026 records for all 19,000 customers. 8,787 terminated-customer monthly usage rows fall after their recorded end month.

• Extreme data usage: 4,480 data usage rows are above the IQR outlier fence; confirm unit and source validity.

• Historical support tickets: ticket dates range from 2014 through 2026; only 622 tickets are dated in 2026. These records are not necessarily contemporaneous with the Q1 2026 usage window.

• Store master cannot be linked to churn or customer geography because no customer-to-store relationship key or customer location attributes were supplied.

• All six files have zero exact duplicate rows. Primary-key uniqueness appears intact for each file's own ID column. This does not by itself prove cross-table referential integrity.

Scope limitations

No plan catalogue, customer demographics, revenue transaction ledger, network outage log, competitor data, or explicit churn event definition was supplied. The analysis therefore cannot compare plan names/features, calculate lifetime value, isolate causal churn drivers, or attribute customer outcomes to stores or network incidents.

6. Recommendations and next steps

Improve subscription data integrity: Standardise subscription_status values, populate missing plan_id values, and validate date chronology and future-dated subscription starts.

Reconcile usage against subscription lifecycle: Investigate Q1 2026 usage rows for customers whose latest termination date is before the usage month. Confirm whether usage rows should be filtered or subscription dates corrected.

Prioritise high-contact customers for retention review: Use repeated support contacts as a risk indicator for further investigation. Trigger a service-recovery review for customers with multiple tickets, but do not assume tickets cause churn.

Improve ticket resolution operations: Review pending, escalated and reopened cases; 42.45% of tickets are not marked Resolved. Track ageing, escalation turnaround and reopened-ticket reasons.

Validate extreme data consumption: Check the 4,480 rows above the IQR threshold and confirm whether the recorded unit is GB, whether values are valid, and whether outliers reflect business customers or data errors.

Build a time-aligned churn model only after validation: Create a defined observation window, target churn event, and features available before the churn date. Use a holdout period and evaluate precision, recall, ROC-AUC and calibration.

Add customer and plan context: Obtain plan catalogue, tenure, geography, device/network quality and billing/payment history to make segmentation and root-cause analysis more actionable.

7. Suggested internship Work Description

Completed the NexaTel Customer Churn Analytics Project using Python, Pandas, NumPy and Excel. Analysed six datasets covering subscription records, voice, SMS and data usage, customer support tickets, and store information. Performed data profiling and quality checks, prepared customer-level subscription and usage summaries, and calculated a status-based churn rate using the latest subscription record for each customer. Examined usage trends, support-ticket resolution metrics, and the association between ticket frequency and subscription status. Identified data-quality issues involving unknown subscription statuses, missing plan IDs, future-dated subscriptions, post-termination usage records, and extreme data-usage values. Presented findings, limitations, and recommendations to improve subscription data quality, customer support operations, and retention analysis.

Submission note

This report is based only on the six CSV files supplied for this project. The results are descriptive and should be refreshed after the identified data-quality issues are resolved. The churn rate is a customer-level status snapshot, not a validated predictive churn model.